ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 9: Some Applications of Trigonometry (Heights and Distances)

Key Concepts & Important Formulas:

* Line of Sight: The line drawn from eye of observer to point in object viewed.

* Angle of Elevation: Angle formed by line of sight with horizontal when object is above horizontal level.

* Angle of Depression: Angle formed by line of sight with horizontal when object is below horizontal level.

* Key Strategy: Form a right angled triangle and apply tan(theta) = Height / Distance.

Solved Exam Examples & Practice:

Example: A tower stands vertically on ground. From a point 15 m away from foot, angle of elevation of top is 60 degrees. Find height of tower.
tan(60) = Height / 15 => sqrt(3) = h / 15 => h = 15*sqrt(3) meters.

Instructor: Mandeep Sir (Green Board) | Video lectures available on Root Classes | Price: FREE